

Lab: Trabajo con AWS Lambda y Computación Serverless

Dificultad: Intermedia |  **Tiempo Estimado:** 60 minutos **Servicios Principales:**  AWS Lambda,  Amazon EC2,  Amazon SNS,  Systems Manager,  EventBridge.

Resumen y Objetivos

En este laboratorio, desplegarás y configurarás una solución de computación **serverless** basada en **AWS Lambda**. La arquitectura automatiza la generación de reportes de ventas diarios extrayendo datos de una base de datos MySQL (alojada en EC2) y enviando los resultados por correo electrónico mediante SNS.

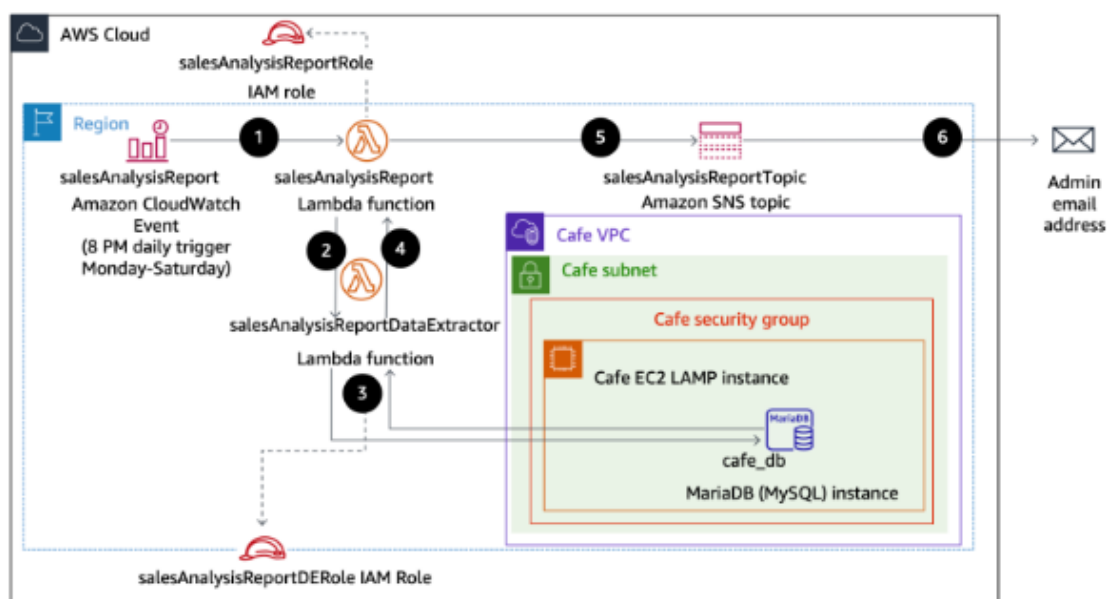
Objetivos técnicos:

-  Verificar permisos de **IAM** para la ejecución de funciones.
-  Crear una **Lambda Layer** para gestionar la librería externa **PyMySQL**.
-  Configurar funciones Lambda para acceder a recursos dentro de una **VPC**.
-  Utilizar la **AWS CLI** para el despliegue de funciones.
-  Automatizar tareas mediante reglas de cron en **Amazon EventBridge**.

Análisis del Escenario

El cliente requiere un sistema de reportes que no consuma recursos de manera ociosa. La solución utiliza una función "Extractora" que interactúa directamente con la base de datos privada en EC2 y una función "Orquestadora" que gestiona la lógica del negocio y las notificaciones. Se utiliza **Systems Manager Parameter Store** para centralizar las credenciales, cumpliendo con el pilar de seguridad de AWS al evitar el "hardcoding" de datos sensibles.

Arquitectura



1 Tarea 1: Observación de permisos IAM

1. **Navega** a la consola de **IAM** y selecciona **Roles**.
2. **Busca** la palabra **sales** en el cuadro de búsqueda para filtrar los roles necesarios.

The screenshot shows the AWS IAM console with the 'Roles' page selected. The search bar at the top contains the text 'Sales', and the results show two roles: 'salesAnalysisReportDERole' and 'salesAnalysisReportRole'. Both roles are trusted by 'AWS Service: lambda'. Below the table, there are sections for 'Roles Anywhere' and 'Access AWS from your non AWS workloads'.

3. **Haz clic** en el rol **salesAnalysisReportRole**.
 - **Verifica** en la pestaña **Trust relationships** que el servicio **lambda.amazonaws.com** aparezca como entidad de confianza.

The screenshot shows the details of the 'salesAnalysisReportRole' in the AWS IAM console. The 'Trust relationships' tab is selected, showing a list of trusted entities. The entity 'lambda.amazonaws.com' is listed with the action 'sts:AssumeRole'.

3. **Analiza** las políticas adjuntas en la pestaña **Permissions**: **AmazonSNSFullAccess**, **AmazonSSMReadOnlyAccess**, **AWSLambdaBasicExecutionRole** y **AWSLambdaRole**.

The screenshot shows the AWS IAM console interface for editing the **AmazonSSMReadOnlyAccess** policy. The page title is "Modify permissions in AmazonSSMReadOnlyAccess". The left sidebar indicates the current step is "Step 1: Modify permissions in AmazonSSMReadOnlyAccess". The main content area is the "Policy editor" in the "JSON" tab. The policy statement is as follows:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Action": [
6         "ssm:Describe*",
7         "ssm:Get*",
8         "ssm:List*"
9       ],
10      "Resource": "*",
11      "Effect": "Allow"
12    }
13  ]
14 }
```

The right sidebar shows the "Add actions" section with a search bar and a list of services. The "Included" section lists "Systems Manager". The "Available" section lists various other services like "AI Operations", "AMP", "API Gateway", etc.

The screenshot shows the AWS IAM console interface for editing the **AWSLambdaBasicRunRole** policy. The page title is "Modify permissions in AWSLambdaBasicRunRole". The left sidebar indicates the current step is "Step 1: Modify permissions in AWSLambdaBasicRunRole". The main content area is the "Policy editor" in the "JSON" tab. The policy statement is as follows:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Action": [
6         "logs:CreateLogGroup",
7         "logs:CreateLogStream",
8         "logs:PutLogEvents"
9       ],
10      "Resource": "*",
11      "Effect": "Allow"
12    }
13  ]
14 }
```

The right sidebar shows the "Add actions" section with a search bar and a list of services. The "Included" section lists "CloudWatch Logs". The "Available" section lists various other services like "AI Operations", "AMP", "API Gateway", etc.

Step 1
 Modify permissions in AWSLambdaRole
 Step 2
 Review and save

Modify permissions in AWSLambdaRole

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Action": [
6         "lambda:InvokeFunction"
7       ],
8       "Resource": [
9         "*"
10      ],
11      "Effect": "allow"
12    }
13  ]
14 }
  
```

Edit statement [Remove](#)

Add actions

Choose a service

Search services

Included
 Lambda

Available
 AI Operations
 AMP
 API Gateway
 API Gateway V2
 ARC Region switch
 ARC Zonal Shift
 ASC
 AWS MCP

Add a resource [Add](#)

Add a condition - optional [Add](#)

[+ Add new statement](#)

7:6 JSON

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4. Regresa a la lista de roles y **selecciona salesAnalysisReportDERole**.

- **Confirma** que incluya la política **AWSLambdaVPCLambdaAccessExecutionRole**. Esta es indispensable para que la función genere interfaces de red (ENI) y acceda a la base de datos dentro de la VPC.

salesAnalysisReportDERole [Info](#) [Delete](#) [Edit](#)

Summary

Creation date
 April 07, 2026, 17:56 (UTC-04:00)

Last activity
 -

ARN
 arn:aws:iam::157676725654:role/salesAnalysisReportDERole

Maximum session duration
 1 hour

Permissions **Trust relationships** **Tags (1)** **Last Accessed** **Revoke sessions**

Trusted entities [Edit trust policy](#)

Entities that can assume this role under specified conditions.

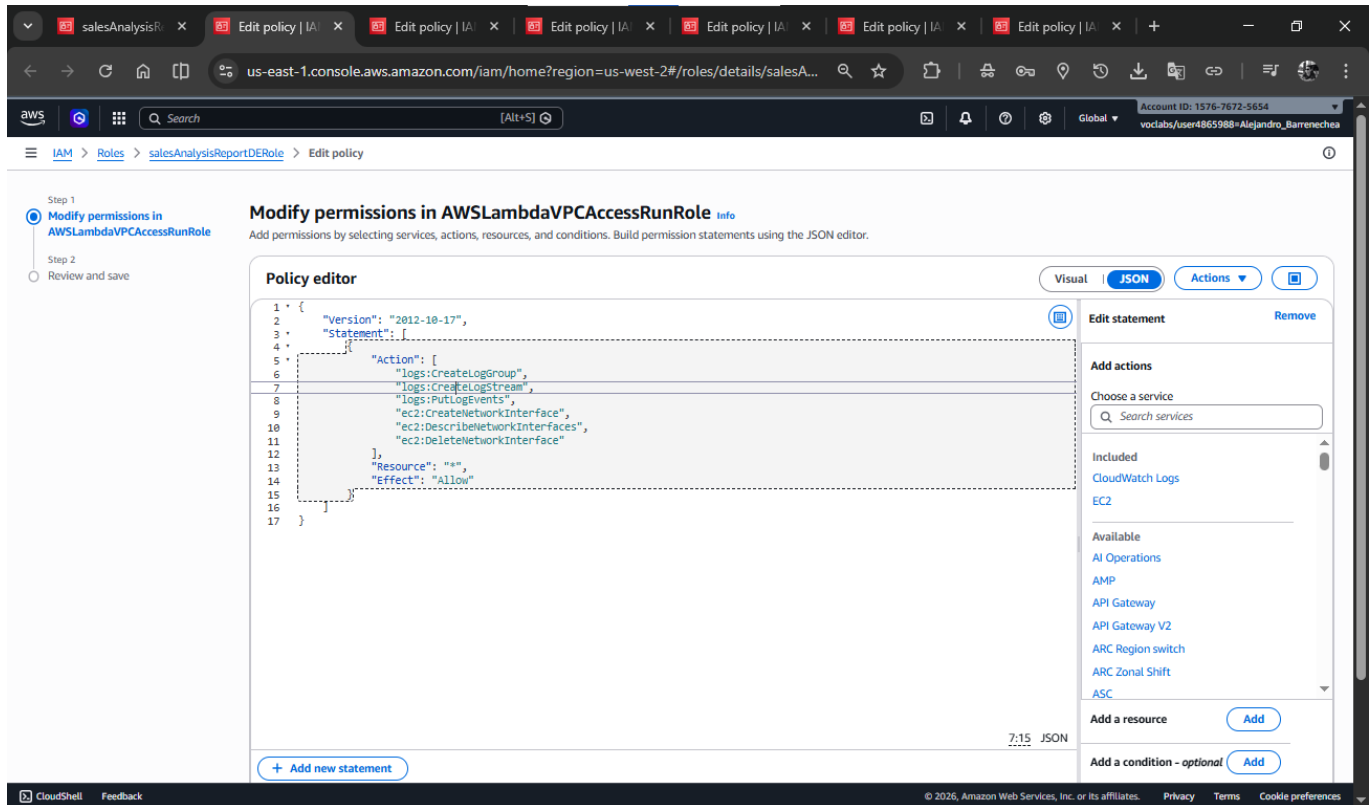
```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "Service": "lambda.amazonaws.com"
8       },
9       "Action": "sts:AssumeRole"
10    }
11  ]
12 }
  
```

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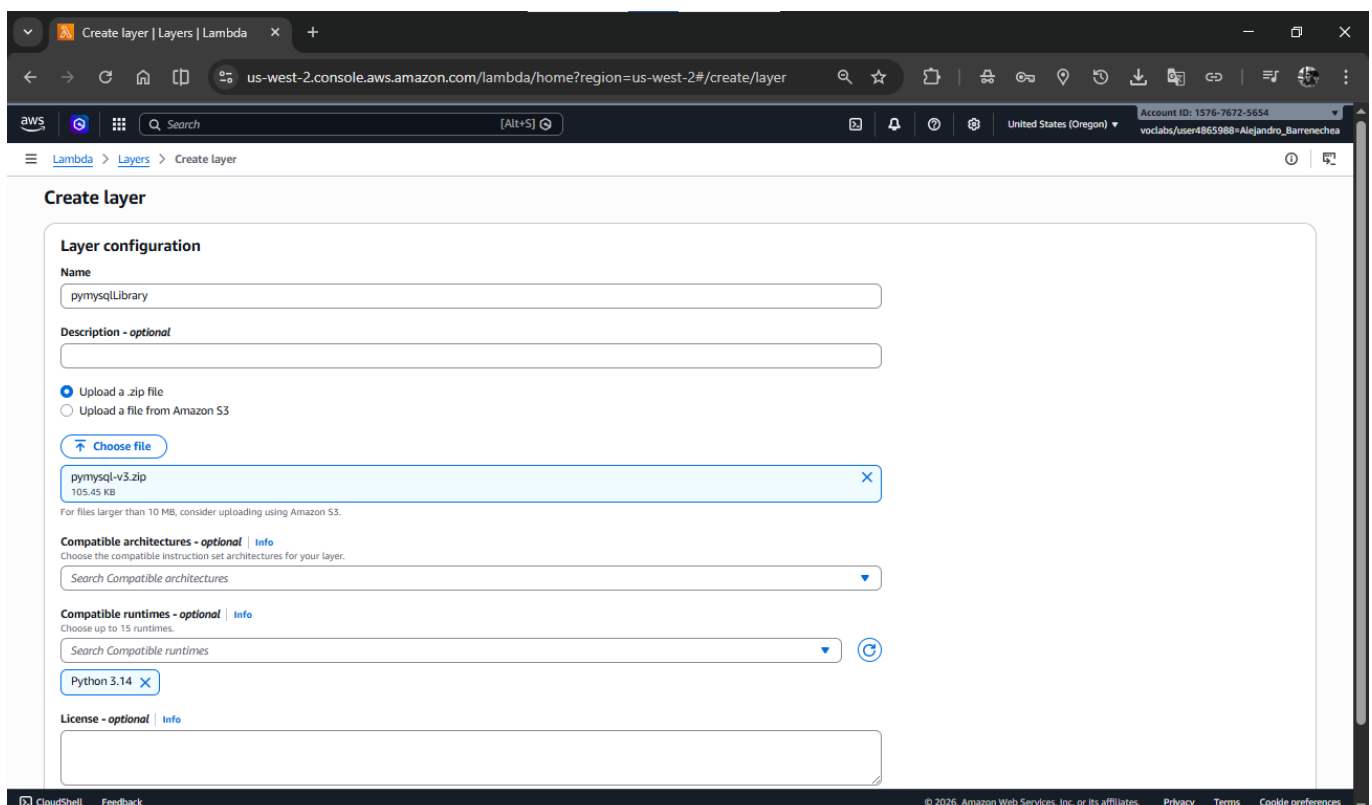
The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, and Access reports. The main content area displays the details for the role **salesAnalysisReportDERole**. The **Summary** section shows the creation date (April 07, 2026, 17:56 UTC-04:00), the ARN (`arn:aws:iam::157676725654:role/salesAnalysisReportDERole`), and the maximum session duration (1 hour). Below the summary, there are tabs for **Permissions**, **Trust relationships**, **Tags (1)**, **Last Accessed**, and **Revoke sessions**. The **Permissions** tab is active, showing a list of **Permissions policies (2)**. The list includes **AWSLambdaBasicRunRole** and **AWSLambdaVPCAccessRunRole**, both of type **Customer inline** and attached to 0 entities. There are buttons for **Simulate**, **Remove**, and **Add permissions**. Below the list, there is a section for **Permissions boundary (not set)** and a section for **Generate policy based on CloudTrail events**.

The screenshot shows the AWS IAM console interface, specifically the **Modify permissions in AWSLambdaBasicRunRole** step. The left sidebar shows the navigation menu. The main content area displays the **Policy editor** for the **AWSLambdaBasicRunRole**. The **Visual** tab is selected, showing a JSON policy document. The **JSON** tab is also visible. The **Visual** tab shows a list of actions: **logs:CreateLogGroup**, **logs:CreateLogStream**, and **logs:PutLogEvents**. The **Resource** is set to **/*** and the **Effect** is **Allow**. The **Visual** tab also shows a list of **Available** actions, including **CloudWatch Logs**, **AI Operations**, **AMP**, **API Gateway**, **API Gateway V2**, **ARC Region switch**, **ARC Zonal Shift**, **ASC**, and **AWS MCP**. There are buttons for **Add new statement**, **Add a resource**, and **Add a condition - optional**.



2 Tarea 2: Creación de Lambda Layer y Función Extractora

1. **Descarga** los archivos: [pymysql-v3.zip](#) y [salesAnalysisReportDataExtractor-v3.zip](#).
2. **Navega** a **AWS Lambda > Layers > Create layer**.
 - **Nombre:** [pymysqlLibrary](#).
 - **Upload:** Sube el archivo [pymysql-v3.zip](#).
 - **Compatible runtimes:** Selecciona **Python 3.14**.
 - **Haz clic en Create.**



3. Crea la función: Ve a **Functions > Create function**.

- **Selecciona:** "Author from scratch".
- **Nombre:** `salesAnalysisReportDataExtractor`.
- **Runtime:** **Python 3.14**.
- **Role:** Selecciona **Use an existing role** y elige `salesAnalysisReportDERole`.
- **Haz clic en Create function.**

Create function

Choose one of the following options to create your function.

- ☒ **Author from scratch**
Start with a simple Hello World example.
- ☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.
- ☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.
`salesAnalysisReportDataExtractor`

Runtime
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.
`Python 3.14`

Durable execution
Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#)
☐ Enable

Architecture
Choose the instruction set architecture you want for your function code.
☒ `x86_64`

Permissions
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

Change default execution role

Execution role
To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

☐ Create default role ☒ Use another role
`salesAnalysisReportDERole` [View role details in IAM](#)

Additional configurations
Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

[Cancel](#) [Create function](#)

4. Añade el Layer: En la sección **Layers** (al final de la página), haz clic en **Add a layer**.

Code properties

Package size: 299 byte
SHA256 hash: `R6TL2F5y9wIPT0t4NqRtY2u043UvWgaXmK4cSHnwRQ=`
Last modified: 6 minutes ago

Runtime settings

Runtime: `Python 3.14`
Handler: `lambda_function.lambda_handler`
Architecture: `x86_64`

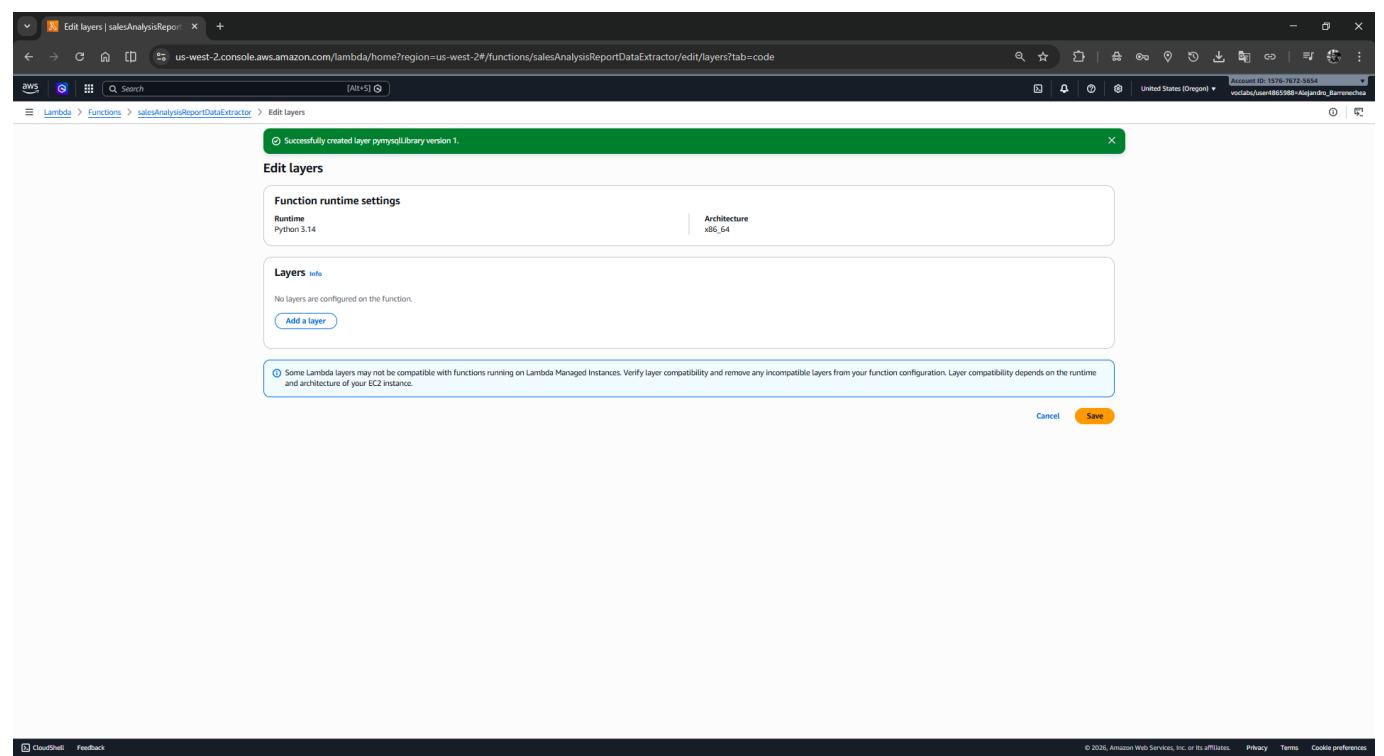
Runtime management configuration

Runtime version ARN: `arn:aws:lambda:us-west-2:runtime:6198f9d9708b48010232459840699b2264ba8c36621c703d5cd51b2245e02d1`
Update runtime version: `Auto`

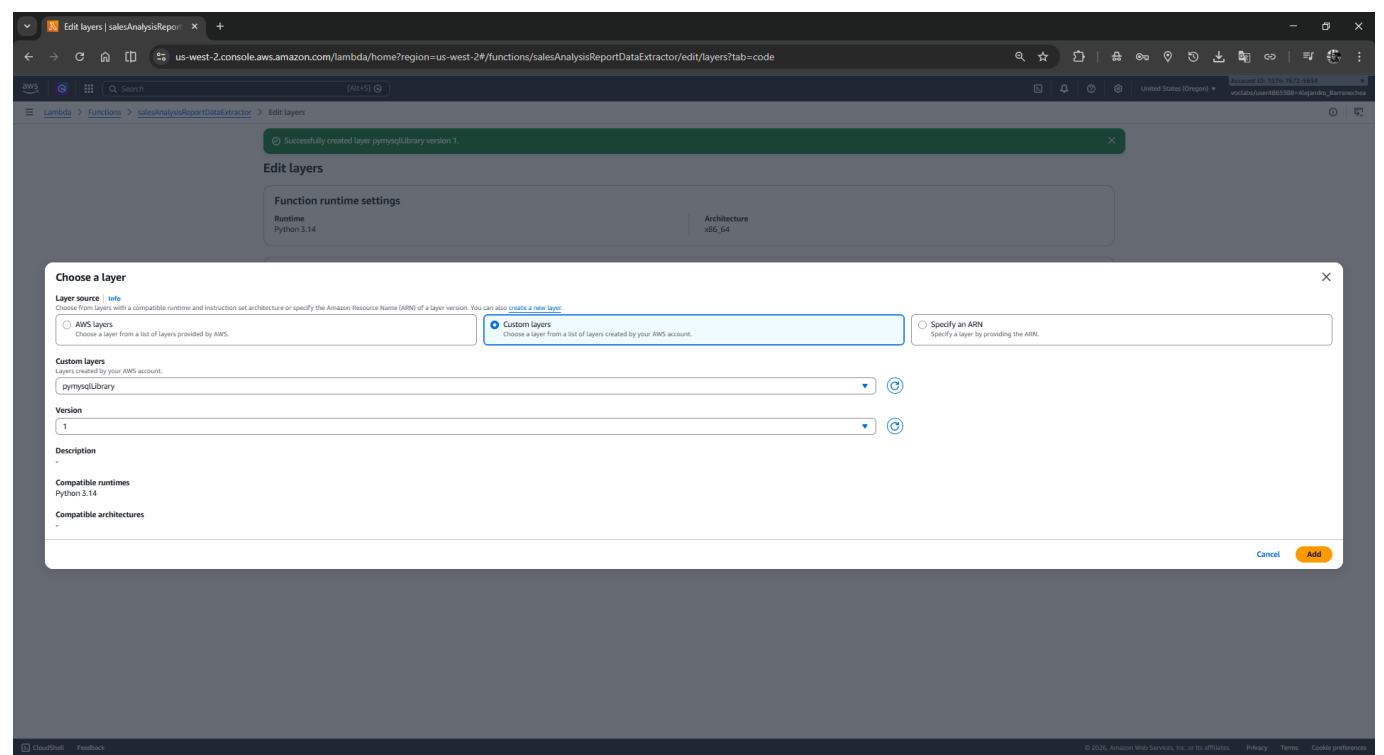
Layers (0)

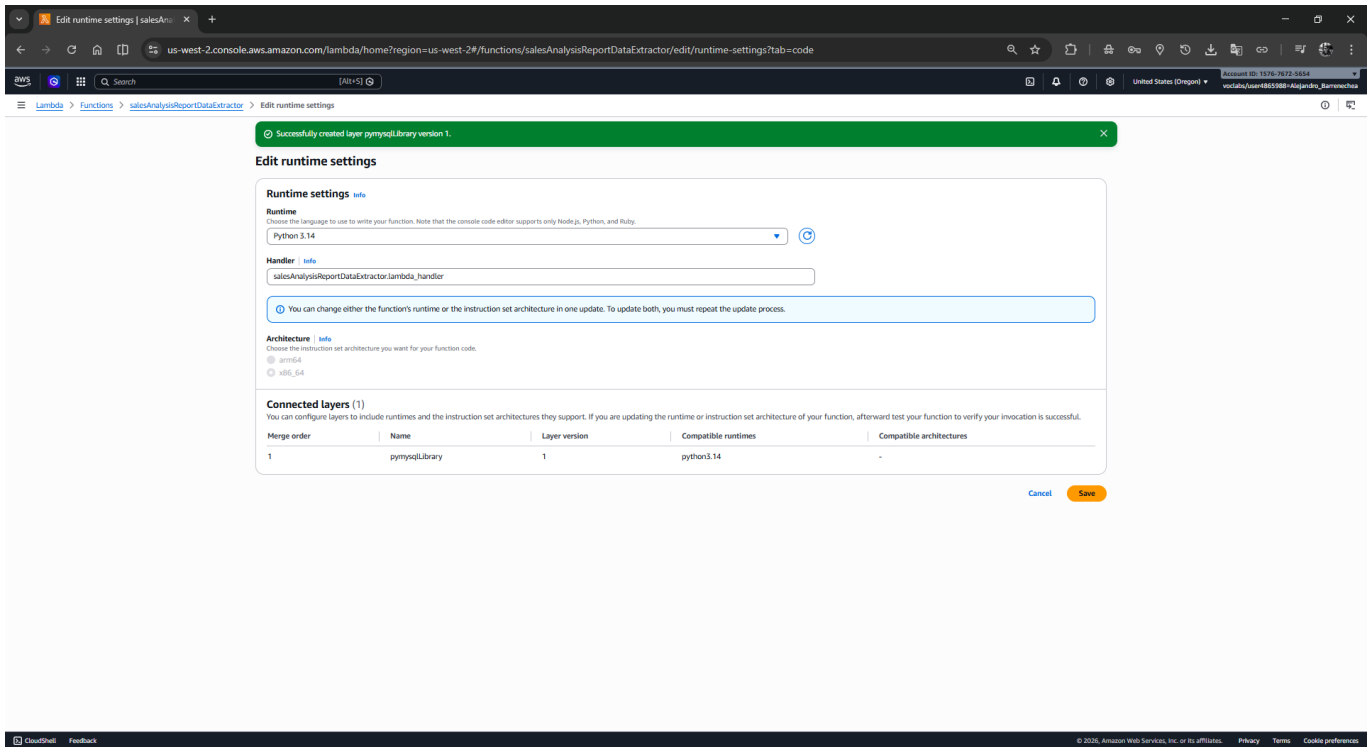
Search by attributes or search by keyword

Merge order	Name	Layer version	Compatible runtimes	Compatible architectures	Version ARN
There is no data to display.					

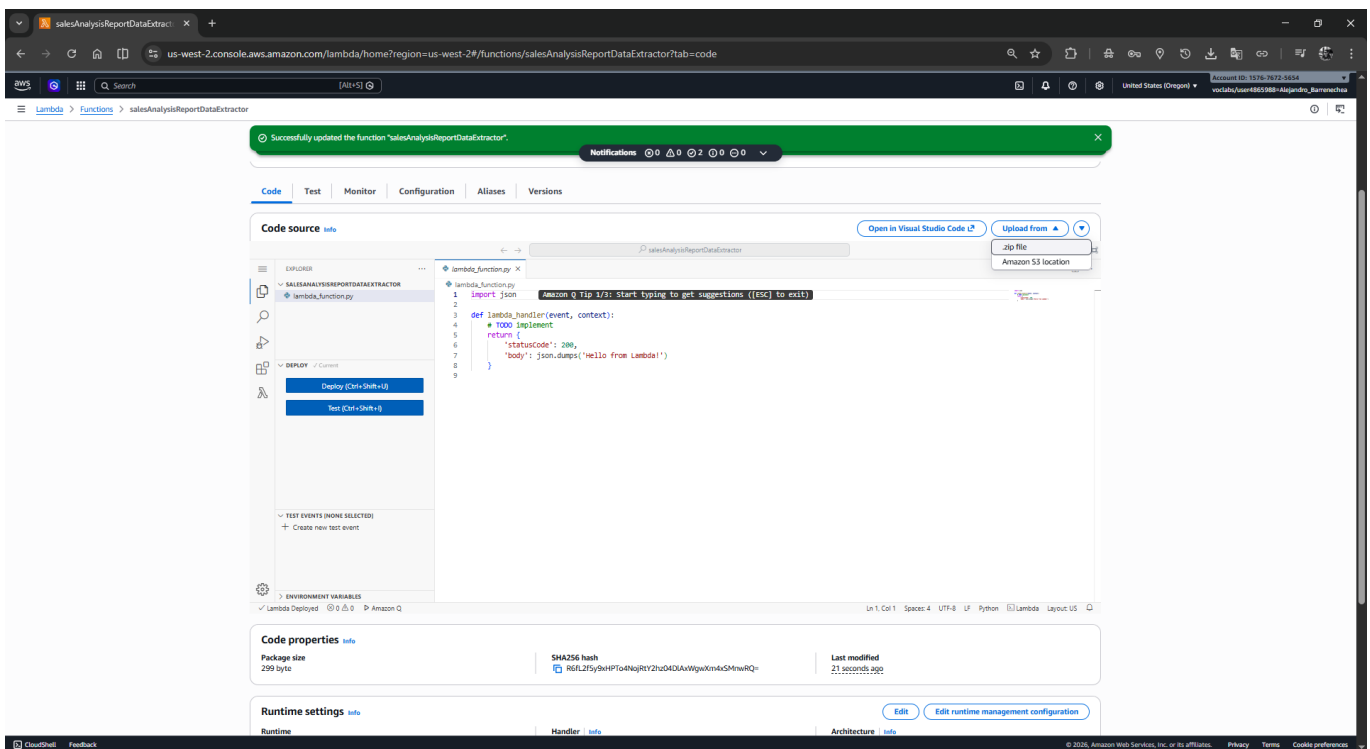


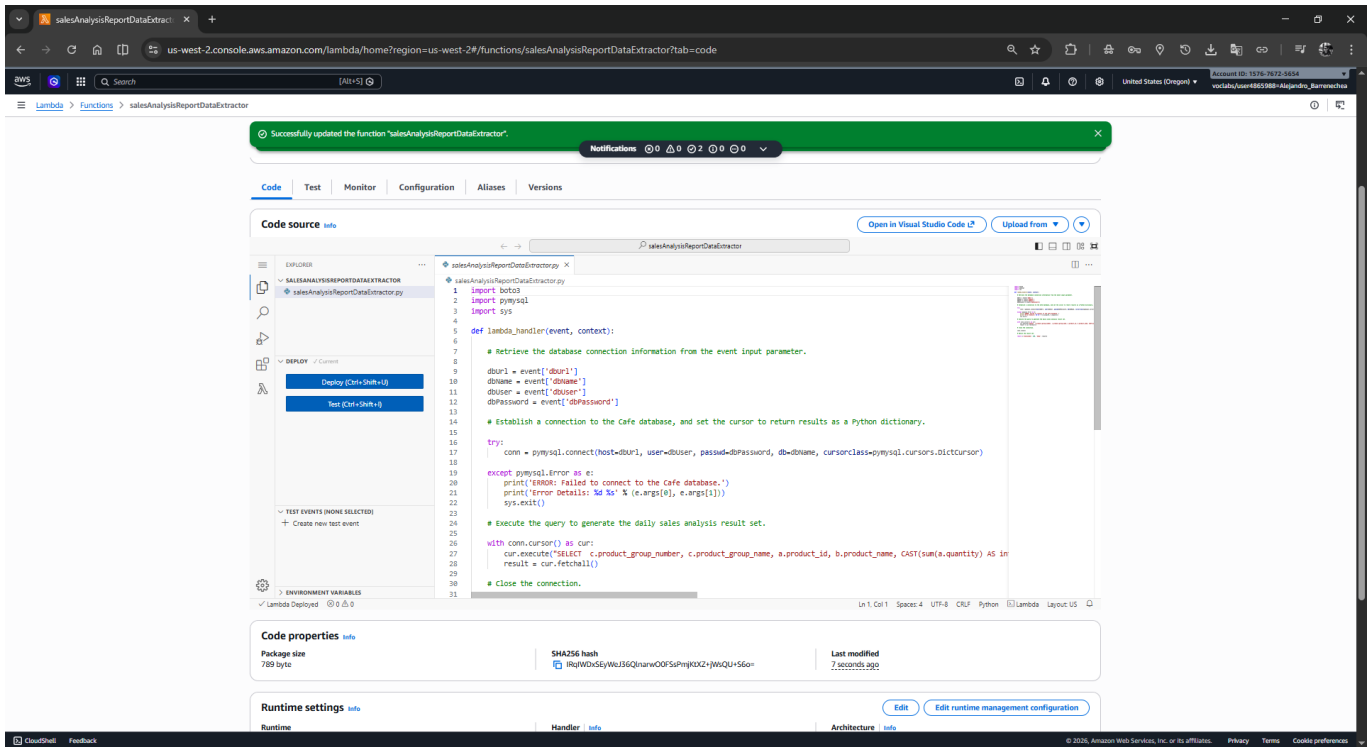
Selecciona: "Custom layers" > pymysqlLibrary > Versión 1.





5. **Importa el código:** En la pestaña **Code**, haz clic en **Upload from** > **.zip file** y carga **salesAnalysisReportDataExtractor-v3.zip**.

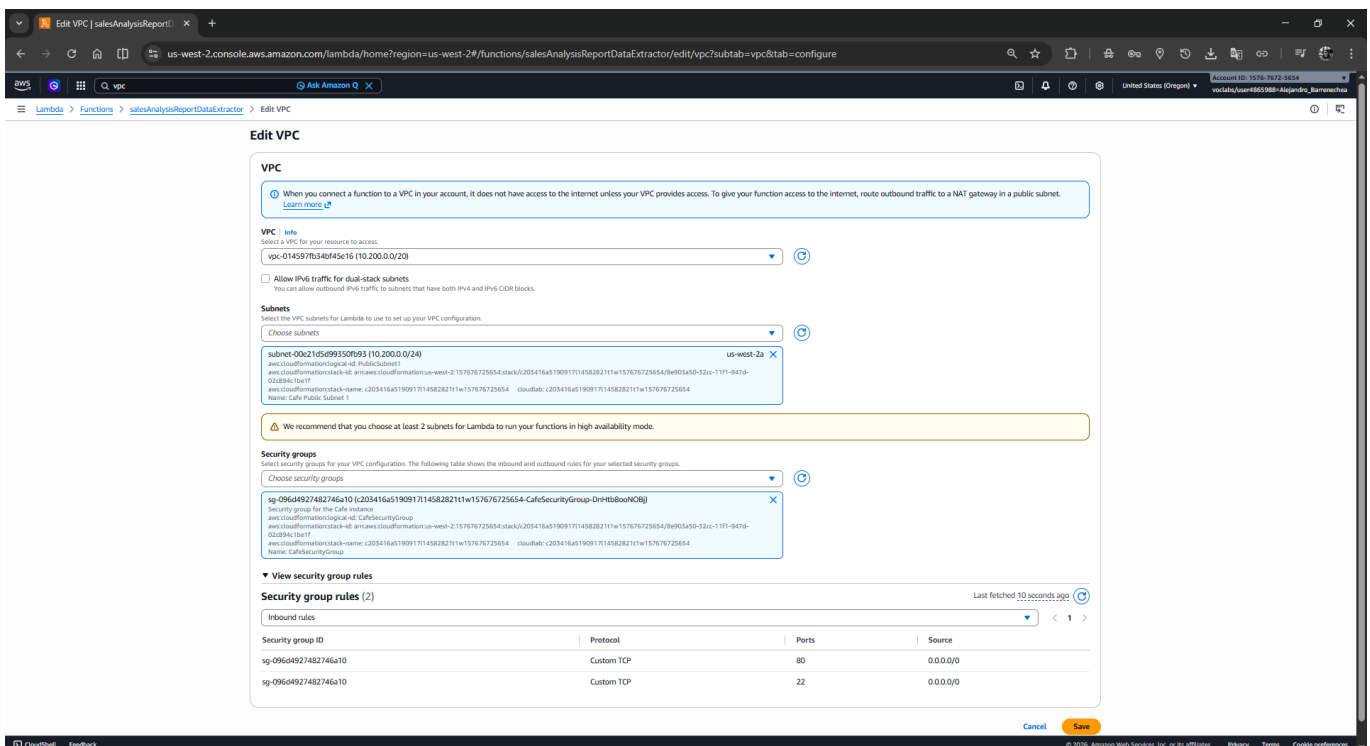




6. **Configura el Handler:** En **Runtime settings**, haz clic en **Edit** y escribe `salesAnalysisReportDataExtractor.lambda_handler`. **Guarda**.

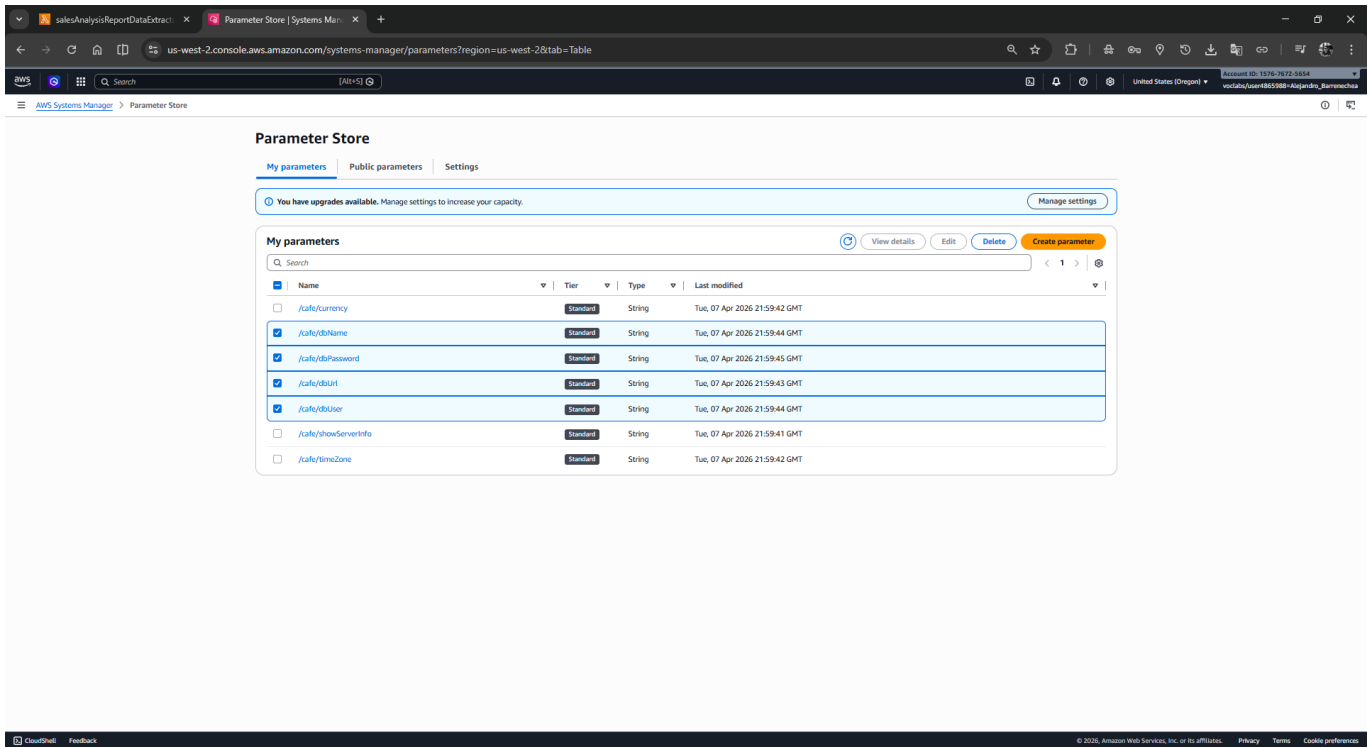
7. **Configura la red:** Ve a **Configuration > VPC > Edit**.

- **VPC:** Selecciona **Cafe VPC**.
- **Subnets:** Elige **Cafe Public Subnet 1**.
- **Security groups:** Selecciona **CafeSecurityGroup**. Haz clic en **Save**.



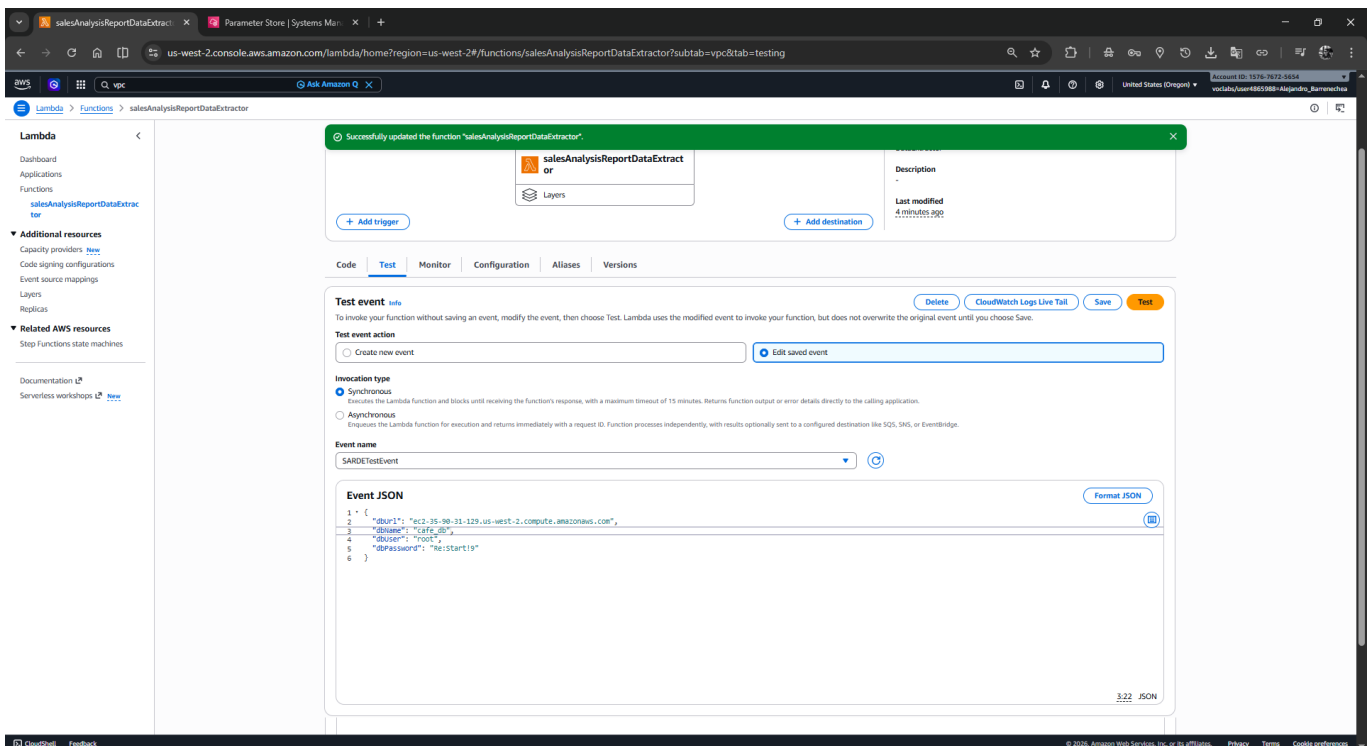
3 Tarea 3: Pruebas y Troubleshooting 🛠️

1. **Obtén parámetros:** Navega a **Systems Manager > Parameter Store**. Copia los valores de `/cafe/dbUrl`, `/cafe/dbName`, `/cafe/dbUser` y `/cafe/dbPassword`.

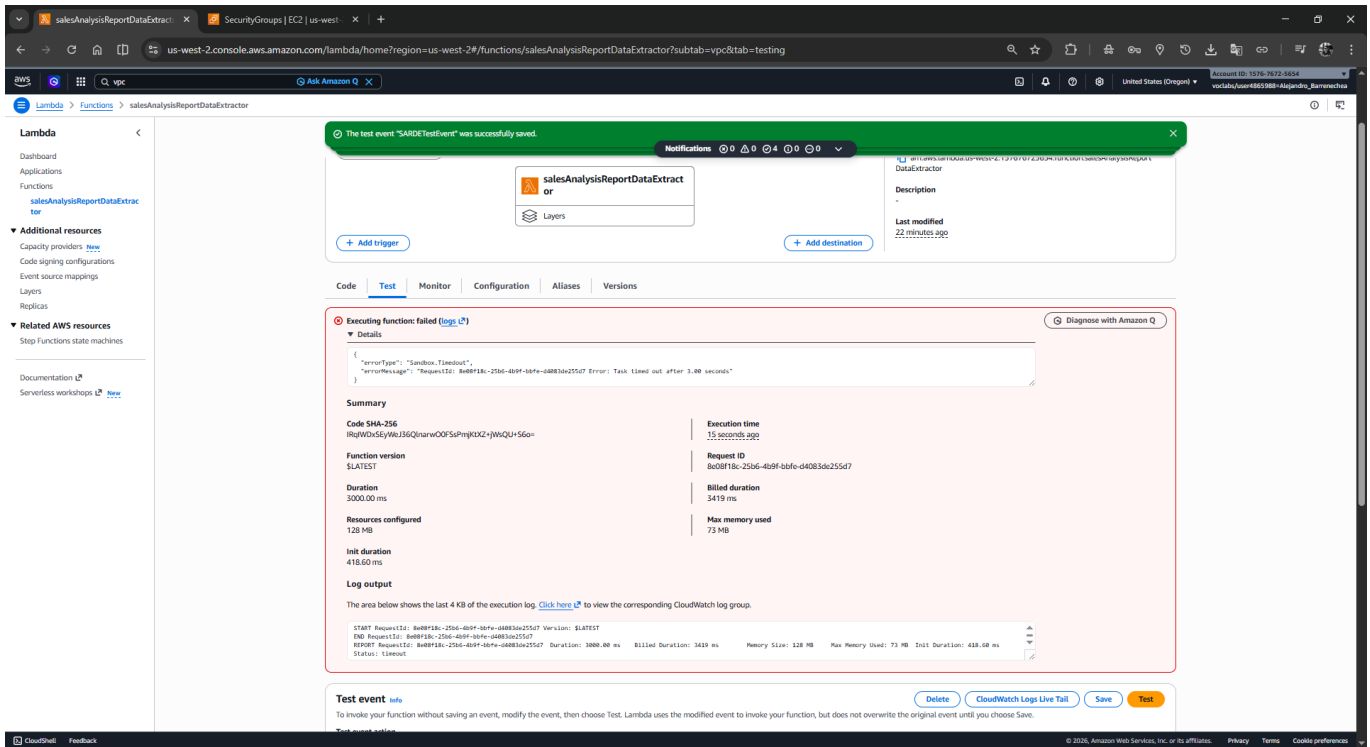


2. Crea el Test: En la pestaña **Test** de tu Lambda.

- Nombre: **SARDETtestEvent**.
- JSON: Reemplaza los valores con los datos reales obtenidos en el paso anterior. **Guarda**.

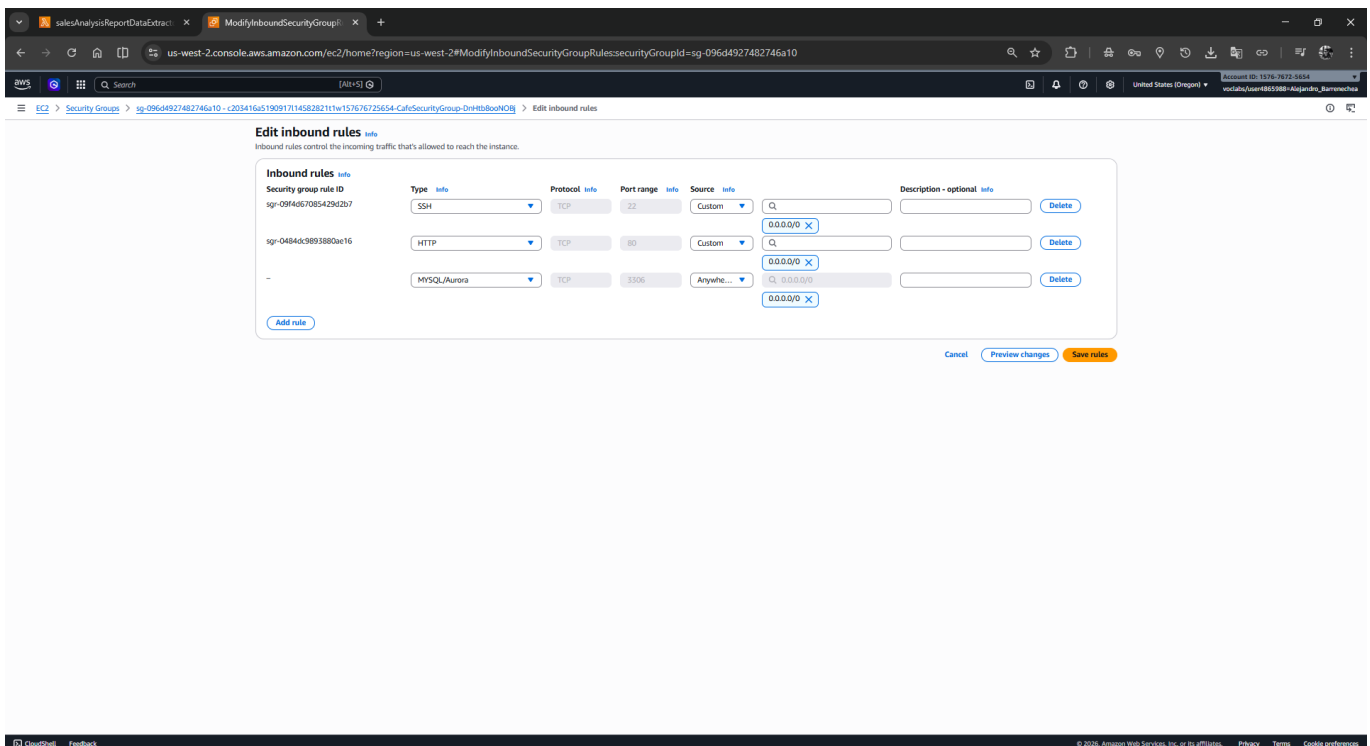


3. Ejecuta el Test: Recibirás un error de **Timeout (3s)**. Este es el comportamiento esperado inicial.

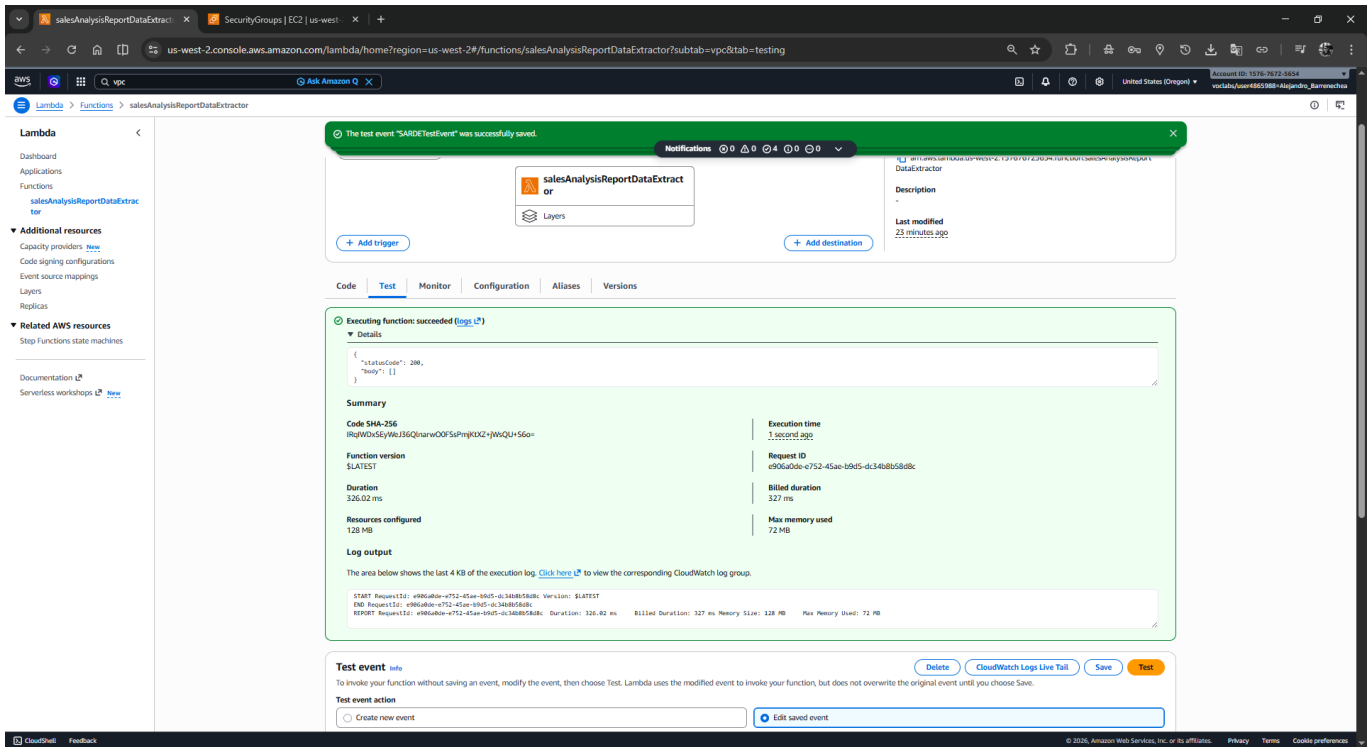


4. Ajuste del Firewall: Ve a **EC2 > Security Groups > CafeSecurityGroup**.

- **Edita las Inbound Rules:** Agrega una regla para **MySQL/Aurora (3306)** permitiendo el tráfico desde **0.0.0.0/0**.

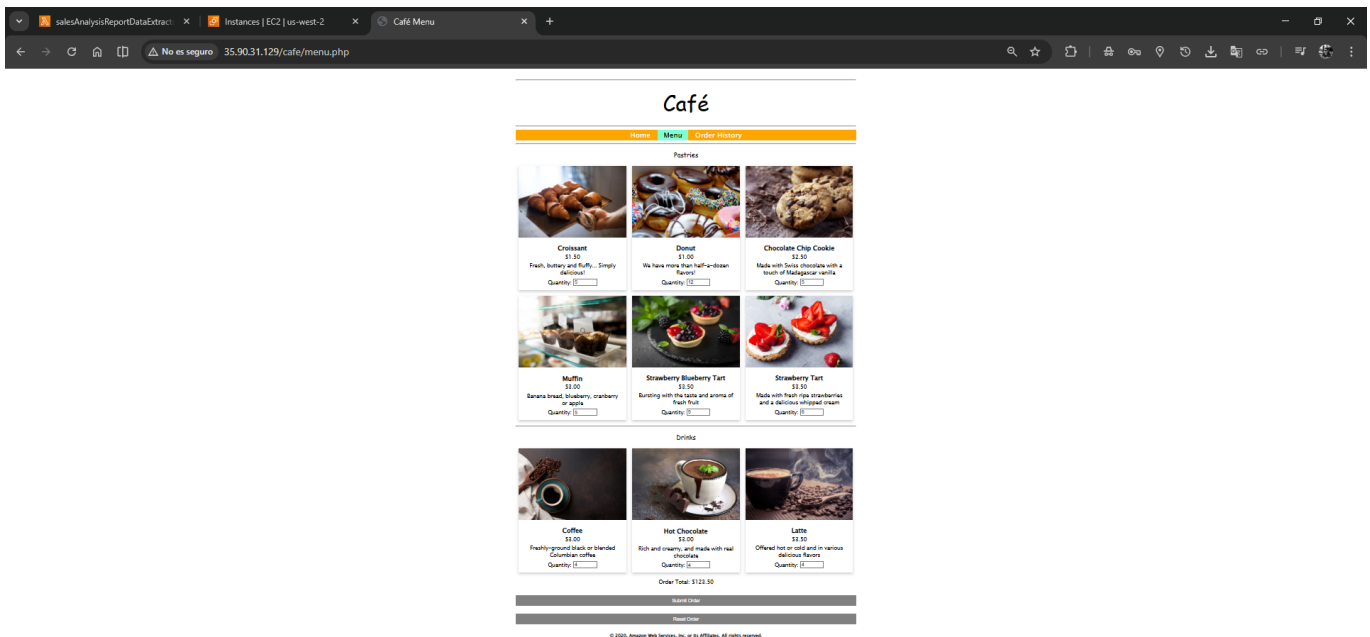


5. Repite el Test: Ahora el resultado debe ser **Succeeded**.

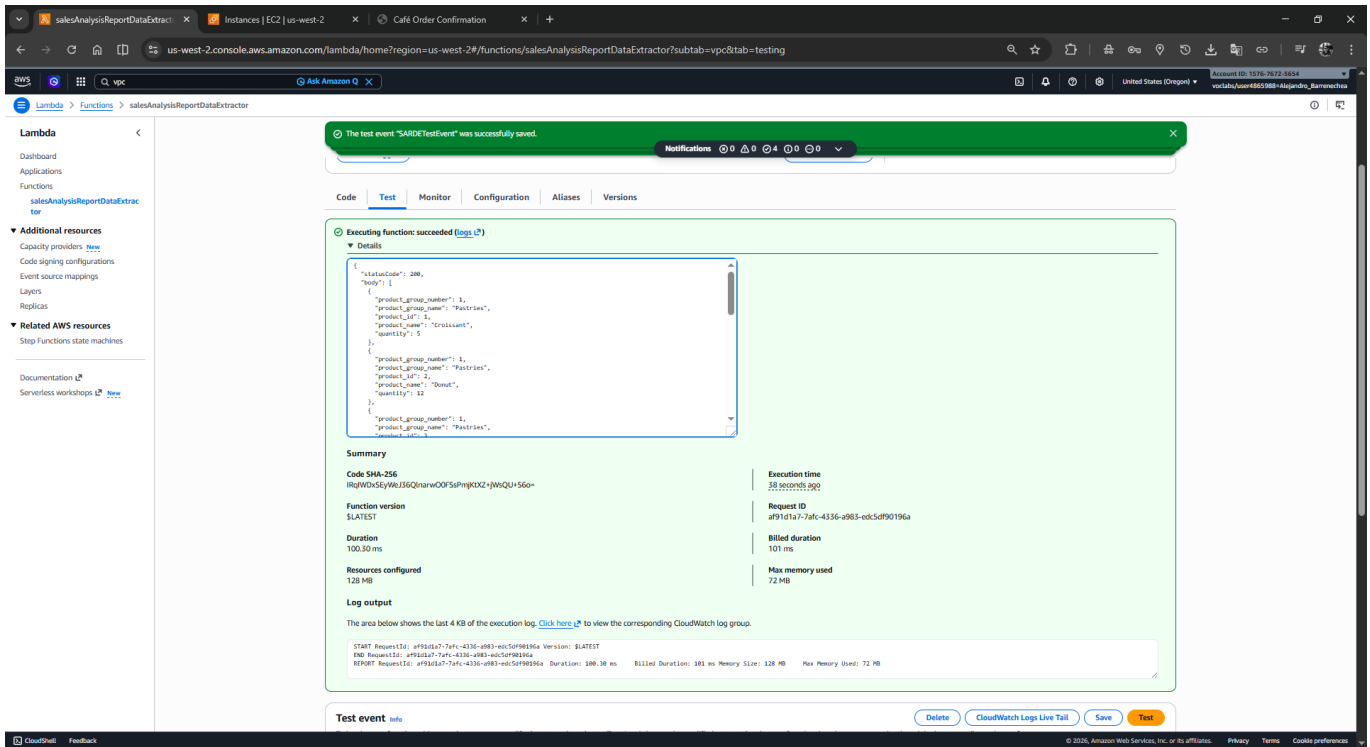


The screenshot shows the AWS Lambda console for the function `salesAnalysisReportDataExtractor`. A notification at the top states: "The test event 'SARDETestEvent' was successfully saved." Below this, the function's configuration is visible, including a description and the last modified time. The 'Test' tab is selected, showing a successful execution event. The event details include a status code of 200 and a body of `{}`. The summary section provides details about the execution, such as the code SHA-256, function version (SLATEST), duration (326.02 ms), and resources configured (128 MB). The log output shows the start and end of the request, along with the duration and billed duration.

6. **Simulación de Negocio:** Accede a la web del café (<http://<PublicIP>/cafe>), realiza pedidos y verifica que la Lambda extraiga los nuevos datos.



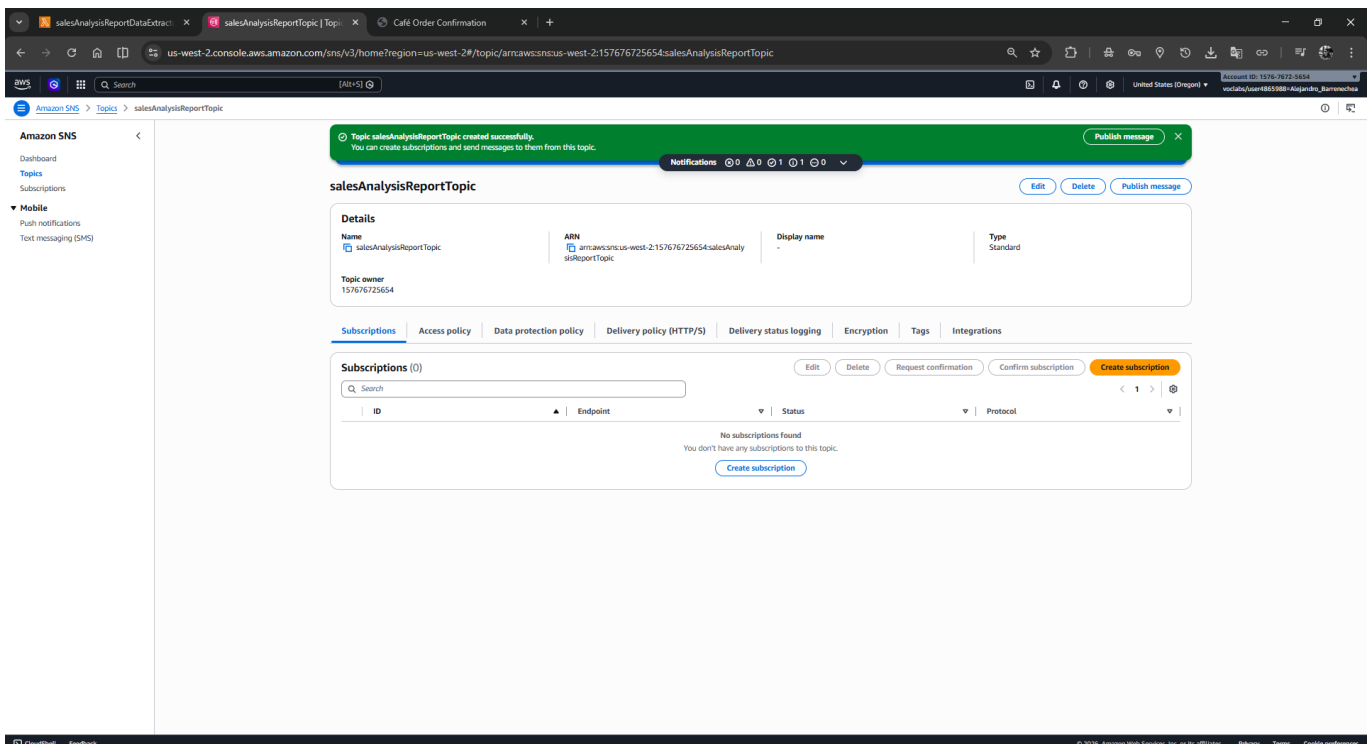
The screenshot shows a web browser displaying the 'Café' menu. The menu is divided into 'Pastries' and 'Drinks' sections. The 'Pastries' section includes items like Croissant, Donut, Chocolate Chip Cookie, Muffin, Strawberry Blueberry Tart, and Strawberry Tart. The 'Drinks' section includes Coffee, Hot Chocolate, and Latte. The order total is \$121.50.



4 Tarea 4: Configuración de Notificaciones (SNS) 🔔

1. Navega a Amazon SNS > Topics > Create topic.

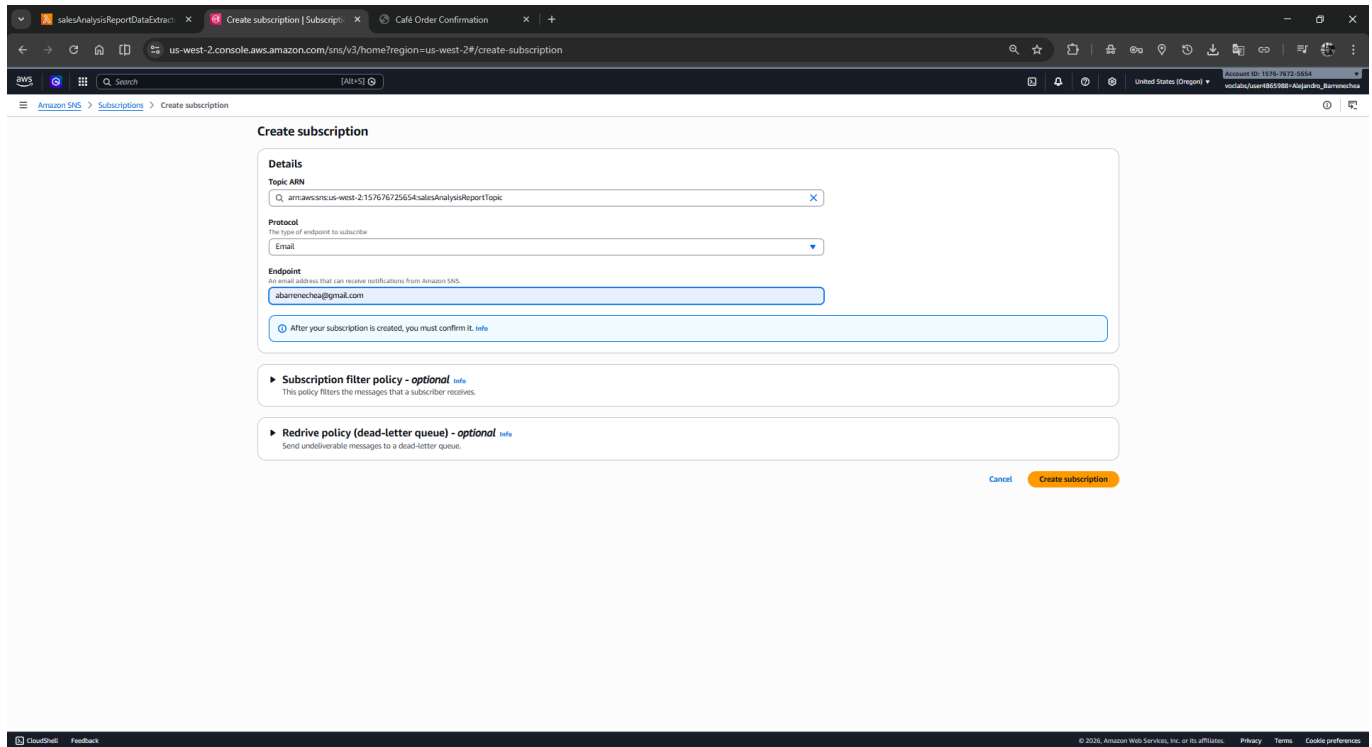
- **Tipo:** Standard. **Nombre:** salesAnalysisReportTopic.



2. Copia el ARN del tópico.

3. Crea la suscripción: Haz clic en **Create subscription**.

- **Protocol:** Email. **Endpoint:** Tu correo electrónico.



Create subscription

Details

Topic ARN
arn:aws:sns:us-west-2:157676725654:salesAnalysisReportTopic

Protocol
Email

Endpoint
abameneche@gmail.com

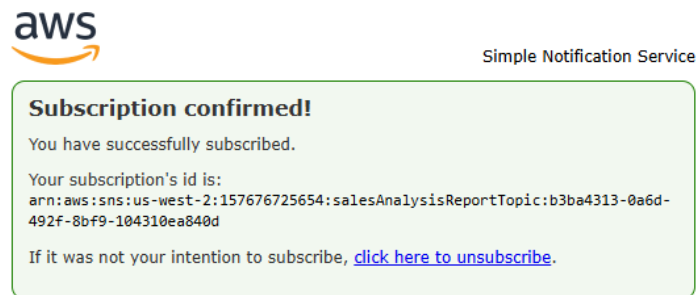
After your subscription is created, you must confirm it. [info](#)

Subscription filter policy - optional [info](#)
This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional [info](#)
Send undeliverable messages to a dead-letter queue.

[Cancel](#) [Create subscription](#)

4. **Confirmación:** Revisa tu email y haz clic en el enlace de confirmación. Asegúrate de ver el mensaje "Subscription confirmed!".



5 Tarea 5: Despliegue con CLI 🖥️

1. **Conéctate:** En EC2, selecciona **CLI Host** y utiliza **EC2 Instance Connect**.

The screenshot shows the AWS Management Console for the 'us-west-2' region. The 'Instances' page displays a table with two instances. The 'CLI Host' instance is selected, and its details are shown on the right. The details include Instance ID, Public IPv4 address, Private IP address, VPC ID, Subnet ID, and Instance ARN.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name
CafeInstance	i-00f240f7b85366b4	Running	t3.micro	3/3 checks pass	View alarms +	us-west-2a	ec2-35-90-31-129.us-w...	35.90.31.129	--	--	disabled	c203416a5190917145...	vockey
CLI Host	i-0fb004ed78c14ac8c	Running	t3.micro	3/3 checks pass	View alarms +	us-west-2a	ec2-34-222-235-11.us...	34.222.235.11	--	--	disabled	c203416a5190917145...	vockey

i-0fb004ed78c14ac8c (CLI Host)

Instance summary

Instance ID: i-0fb004ed78c14ac8c

Public IPv4 address: 34.222.235.11 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-10-200-0-240.us-west-2.compute.internal

Instance type: t3.micro

VPC ID: vpc-014597b34b45e16 (Cafe Public VPC)

Subnet ID: subnet-00c21d5d99350fb93 (Cafe Public Subnet 1)

Instance ARN: arn:aws:ec2:us-west-2:157676725654:instance/i-0fb004ed78c14ac8c

Instance details

AMI ID: ami-0a37c0e770b254a5

AMI name: ami2-ami-hvm-2.0.20260330.0-486_64-gp2

Monitoring: disabled

Allowed image: Linux/amd64

Platform details: Linux/UNIX

Termination protection: Disabled

Managed: false

The screenshot shows the 'Connect to instance' page in the AWS Management Console. The 'Connect' tab is active, displaying fields for Instance ID, VPC ID, Security groups, and IAM role. The 'EC2 Instance Connect' section shows the 'Public IPv4 address' and 'Public IPv6 address' fields, along with the 'Username' field.

Connect

Instance ID: i-0fb004ed78c14ac8c (CLI Host)

VPC ID: vpc-014597b34b45e16

Security groups: sg-065f3a0cc702773c (c203416a51909171458282117e157676725654-CLIHostInstanceSG-FjdJ3yAvm6G)

IAM role: --

EC2 Instance Connect

Instance ID: i-0fb004ed78c14ac8c (CLI Host)

Connection type: ☒ Connect using a Public IP (Connect using a public IPv4 or IPv6 address)

☐ Connect using a Private IP (Connect using a private IP address and a VPC endpoint)

Public IPv4 address: 34.222.235.11

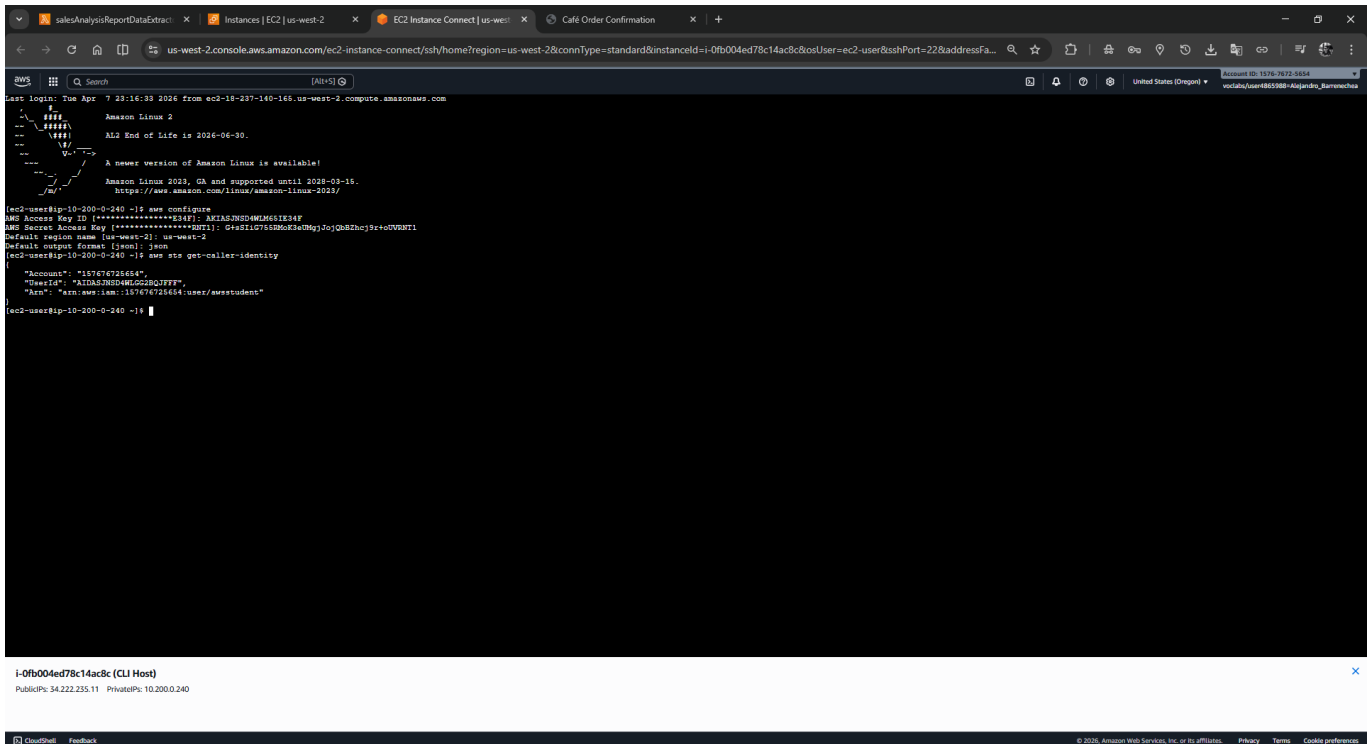
Public IPv6 address: --

Username: ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect

2. **Configura el CLI:** Ejecuta `aws configure` e ingresa tu **Access Key**, **Secret Key** y la región **us-west-2**.



```

Last login: Tue Apr 7 23:16:33 2026 from ec2-18-237-140-146.us-west-2.compute.amazonaws.com
Amazon Linux 2
--
#####
--  \
--  / End of Life is 2026-06-30.
--  /
--  \
--  / A newer version of Amazon Linux is available!
--  \
--  / Amazon Linux 2023, GA and supported until 2028-03-15.
--  \
--  / https://aws.amazon.com/linux/amazon-linux-2023/
--  \
[ec2-user@ip-10-200-0-240 ~]$ aws configure
AWS Access Key ID [*****]: AKIAS3NSD4WJG2JQ2FFFF
AWS Secret Access Key [*****]: Gz310r58Moc3eMg1JoQb2hcj9rtoUVEWT1
Default region name [us-west-2]: us-west-2
Default output format [json]: json
[ec2-user@ip-10-200-0-240 ~]$ aws sts get-caller-identity
{
  "Account": "157676725654",
  "UserId": "AKIAS3NSD4WJG2JQ2FFFF",
  "Arn": "arn:aws:iam::157676725654:user/awstudent"
}
[ec2-user@ip-10-200-0-240 ~]$

```

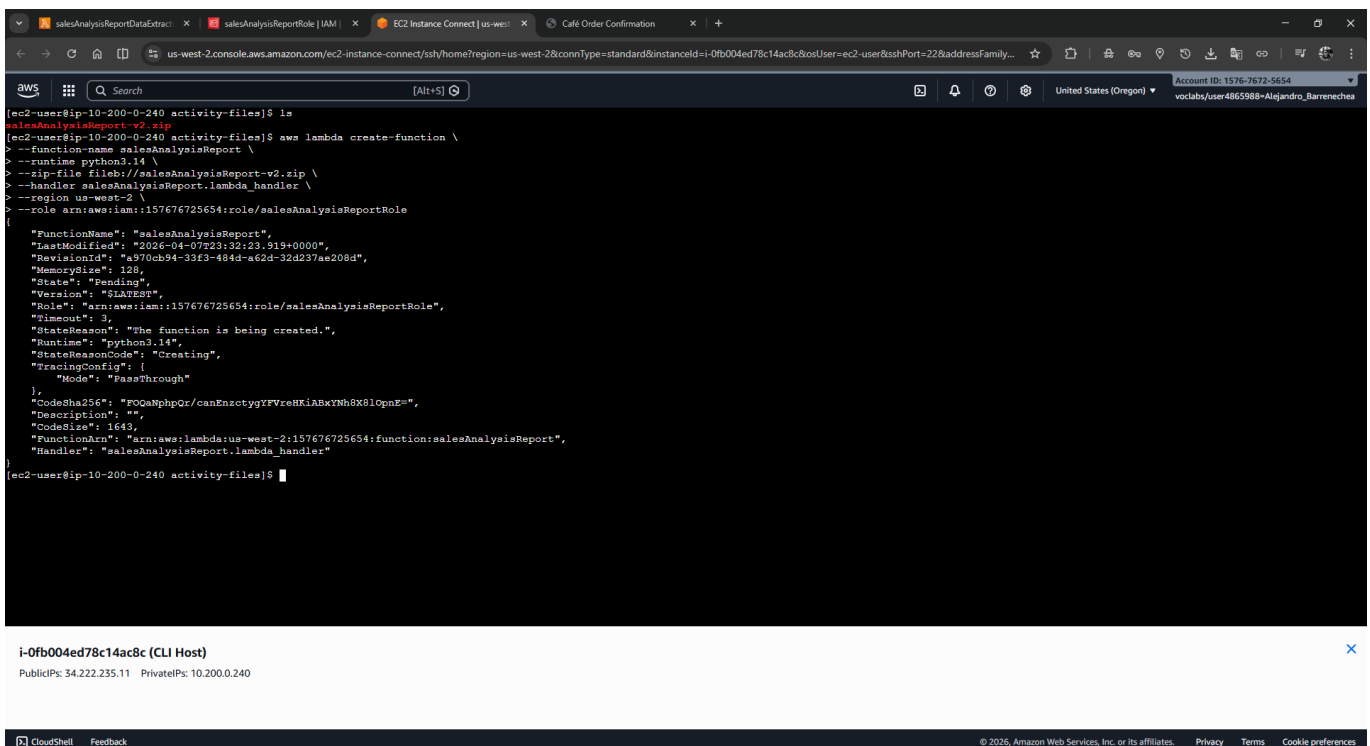
i-0fb004ed78c14ac8c (CLI Host)
PublicIPs: 34.222.235.11 PrivateIPs: 10.200.0.240

3. Crea la función orquestadora:

```

aws lambda create-function \
--function-name salesAnalysisReport \
--runtime python3.14 \
--zip-file fileb://salesAnalysisReport-v2.zip \
--handler salesAnalysisReport.lambda_handler \
--region us-west-2 \
--role <ARN_DEL_ROL_IAM_SALES_ANALYSIS_REPORT>

```



```

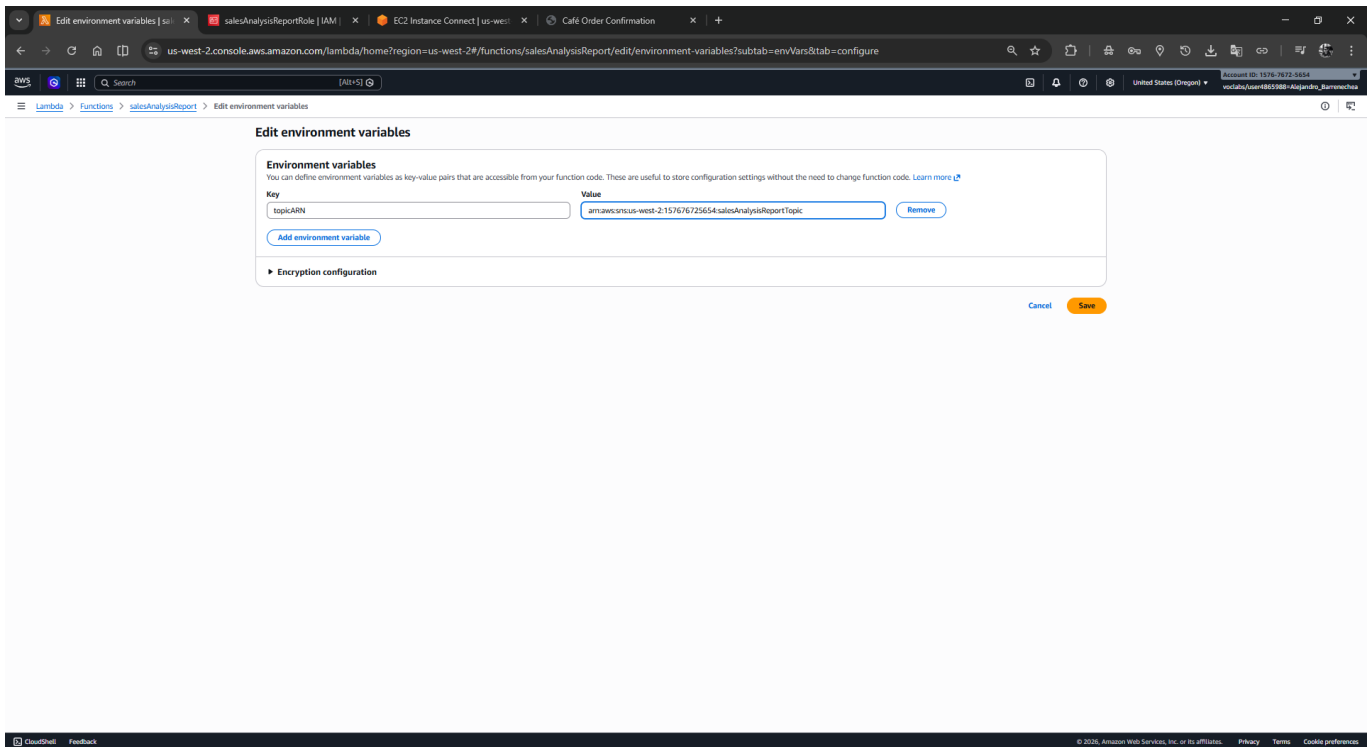
[ec2-user@ip-10-200-0-240 activity-files]$ ls
salesAnalysisReport-v2.zip
[ec2-user@ip-10-200-0-240 activity-files]$ aws lambda create-function \
> --function-name salesAnalysisReport \
> --runtime python3.14 \
> --zip-file fileb://salesAnalysisReport-v2.zip \
> --handler salesAnalysisReport.lambda_handler \
> --region us-west-2 \
> --role arn:aws:iam::157676725654:role/salesAnalysisReportRole
{
  "FunctionName": "salesAnalysisReport",
  "LastModified": "2026-04-07T23:32:23.919+0000",
  "RevisionId": "a970cb94-33f3-484d-a62d-32d237ae208d",
  "MemorySize": 128,
  "State": "Pending",
  "Version": "$LATEST",
  "Role": "arn:aws:iam::157676725654:role/salesAnalysisReportRole",
  "Timeout": 3,
  "StateReason": "The function is being created.",
  "Runtime": "python3.14",
  "StateReasonCode": "Creating",
  "TracingConfig": {
    "Mode": "PassThrough"
  },
  "CodeSha256": "FOGaNphpQr/canEncsctygyYFvrehK1ABxYnH8XB1OpnE=",
  "Description": "",
  "CodeSize": 1643,
  "FunctionArn": "arn:aws:lambda:us-west-2:157676725654:function:salesAnalysisReport",
  "Handler": "salesAnalysisReport.lambda_handler"
}
[ec2-user@ip-10-200-0-240 activity-files]$

```

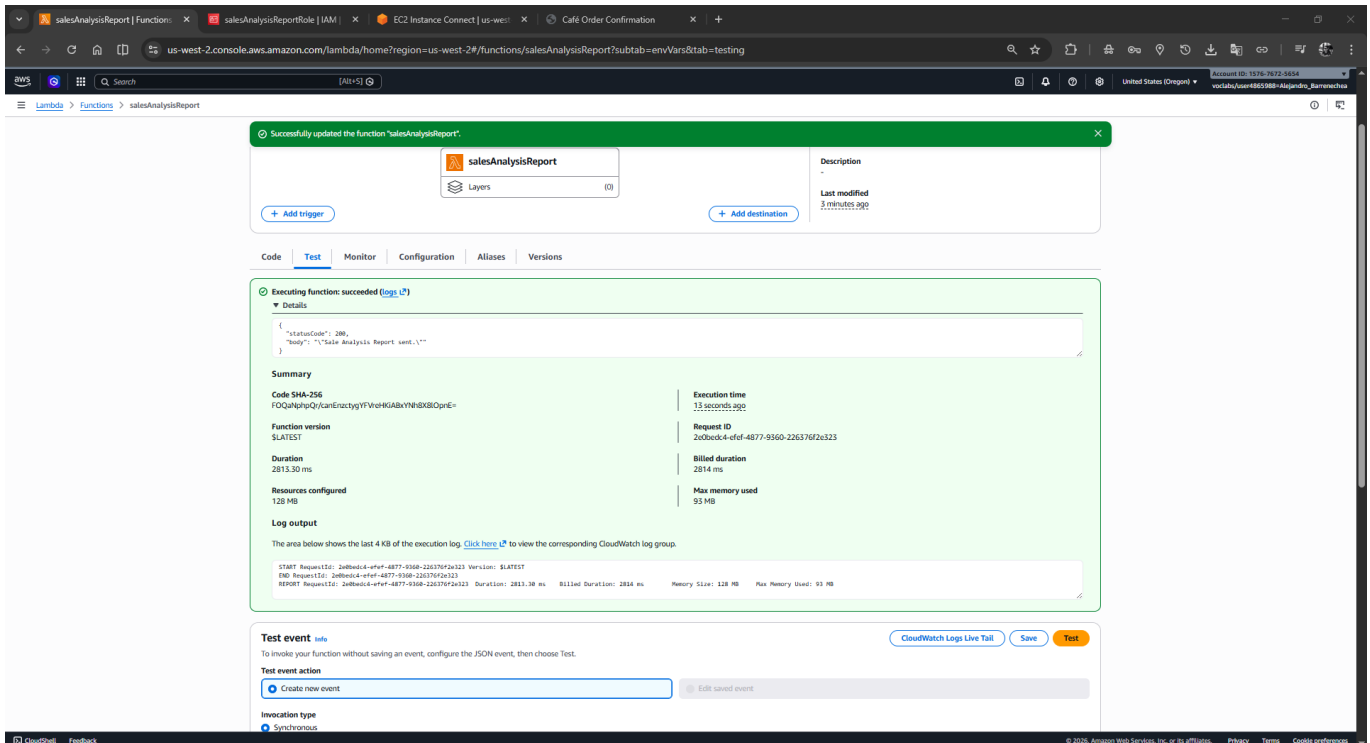
i-0fb004ed78c14ac8c (CLI Host)
PublicIPs: 34.222.235.11 PrivateIPs: 10.200.0.240

4. Variables de entorno: En la consola de Lambda > **salesAnalysisReport** > **Configuration** > **Environment variables**.

- **Añade:** Key: **topicARN** | Value: El ARN de tu tópico SNS.



5. Test Final: Ejecuta **SARTestEvent**. Recibirás el reporte final en tu correo.



6 Tarea 6: Automatización (EventBridge)

1. En la función **salesAnalysisReport**, **añade un trigger** de **EventBridge**.
2. **Crea una nueva regla:** **salesAnalysisReportDailyTrigger**.
3. **Expresión Cron:** Para ejecutar el reporte, usa por ejemplo **cron(0 20 ? * MON-SAT *)** para las 8 PM.

Respuestas Analíticas

- **¿Por qué usamos Python 3.14?** Al ser el runtime recomendado por la consola moderna, garantiza el acceso a las últimas optimizaciones de rendimiento y soporte extendido de AWS.
- **¿Cuál es la función del Security Group en este lab?** Funciona como un firewall virtual que permite que el tráfico MySQL (puerto 3306) originado en la interfaz de red de la Lambda llegue al servidor EC2. Sin esta regla, la conexión se bloquea.
- **¿Qué ventaja ofrece Parameter Store?** Permite que los datos de conexión sean variables. Si la base de datos se migra a otro servidor, solo actualizamos el parámetro en Systems Manager y la arquitectura Lambda se adapta instantáneamente sin tocar el código.